TRAFFIC ACCIDENT SEVERITY PREDICTION

Project Id:20

Name : Anshika Gaur (Team Leader)

Class Roll Number: 1

University Roll Number : 2027616

Section and Group: G(G1)

Name : Anushka Jain

Class Roll Number: 33

University Roll Number : 2028655

Section and Group: G(G1)

1. **Problem Description** 3
2. **Architecture** 4
3. **Models model** 5
4. **Evaluation metrics used** 6
5. **Preliminary Results and Visualizations** 7
6. **References** 8

1. Problem Description

The **Traffic Accident Severity Prediction System** is a machine learning-based application developed to predict the severity of road accidents using factors such as speed, weather conditions, road conditions, and vehicle type. The system aims to assist in identifying high-risk scenarios and improving road safety through data-driven insights.

The dataset is preprocessed by handling missing values and removing irrelevant records to improve data quality. Feature selection techniques are applied to retain the most important attributes, such as speed and weather conditions, which significantly influence accident severity.

2. Architecture

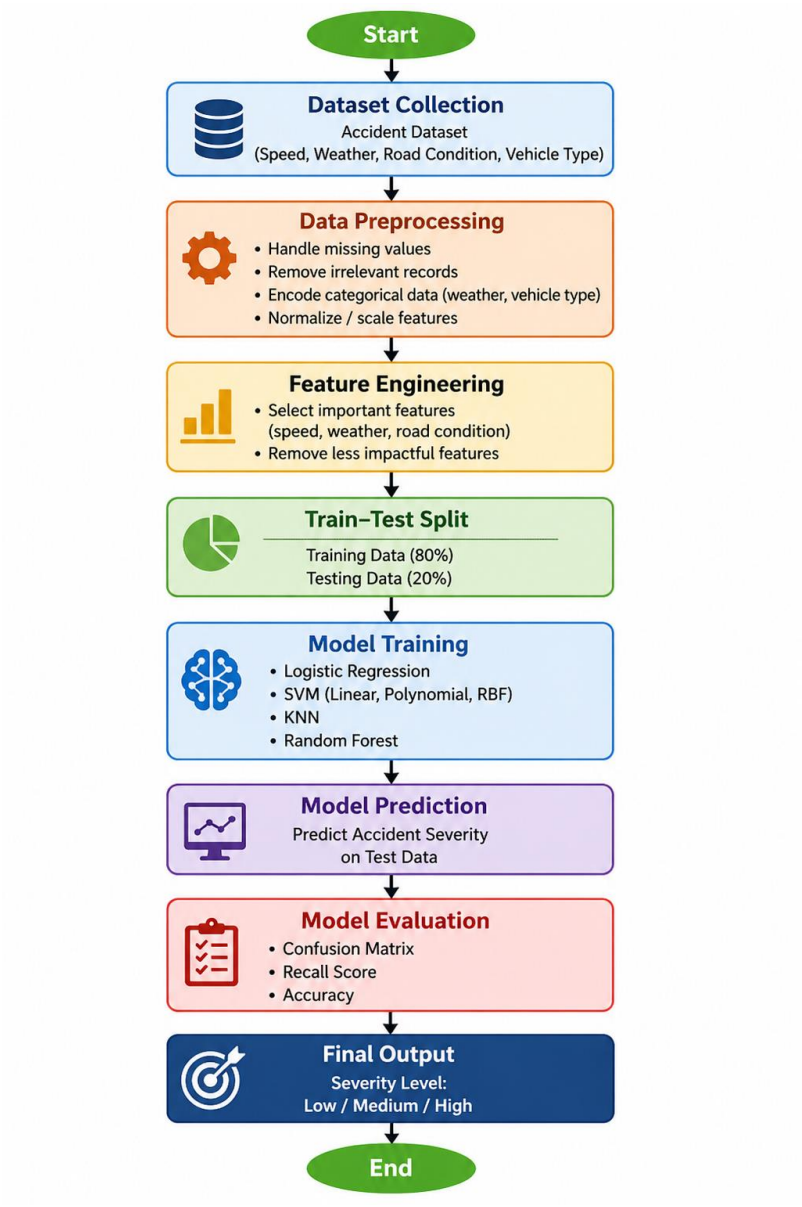

Fig.1 “Traffic Accident Severity Prediction Workflow”

3. Models Used(Applied/ Planned)

- **Logistic Regression**
  Used for basic classification of accident severity. It performs well for linearly separable data.

- **Support Vector Machine (SVM)**
  Applied with different kernel functions:
  - Linear Kernel – for simple data separation
  - Polynomial Kernel – for curved decision boundaries
  - RBF Kernel – for complex and non-linear patterns

- **K-Nearest Neighbors (KNN)**
  Classifies accident severity based on similarity with nearest data points. Its performance depends on the value of K.

- **Random Forest Classifier**
  An ensemble learning method that uses multiple decision trees to improve accuracy and reduce overfitting. It performs well on complex datasets.

4. Evaluation metrics used

- **Confusion Matrix**
  Used to compare actual and predicted values and evaluate model performance.

- **Recall Score**
  Measures the ability of the model to correctly identify severe accident cases, which is important in safety-related applications.
- **Accuracy**
  Used to compare overall performance of different models.

5. Preliminary Results and Visualizations

- Data preprocessing improved dataset quality by removing noise and missing values.
- Feature selection identified **speed and weather conditions** as major contributing factors.
- Multiple models (Logistic Regression, SVM, KNN, Random Forest) were trained and evaluated.

Observations:

- **Random Forest** achieved better accuracy due to ensemble learning.
- **SVM (RBF kernel)** performed well for complex patterns.
- **KNN** gave moderate performance and depends on value of K.
- **Logistic Regression** worked well for simple relationships but less effective for complex data.
- Recall score analysis showed better detection of severe accidents in Random Forest and SVM models.

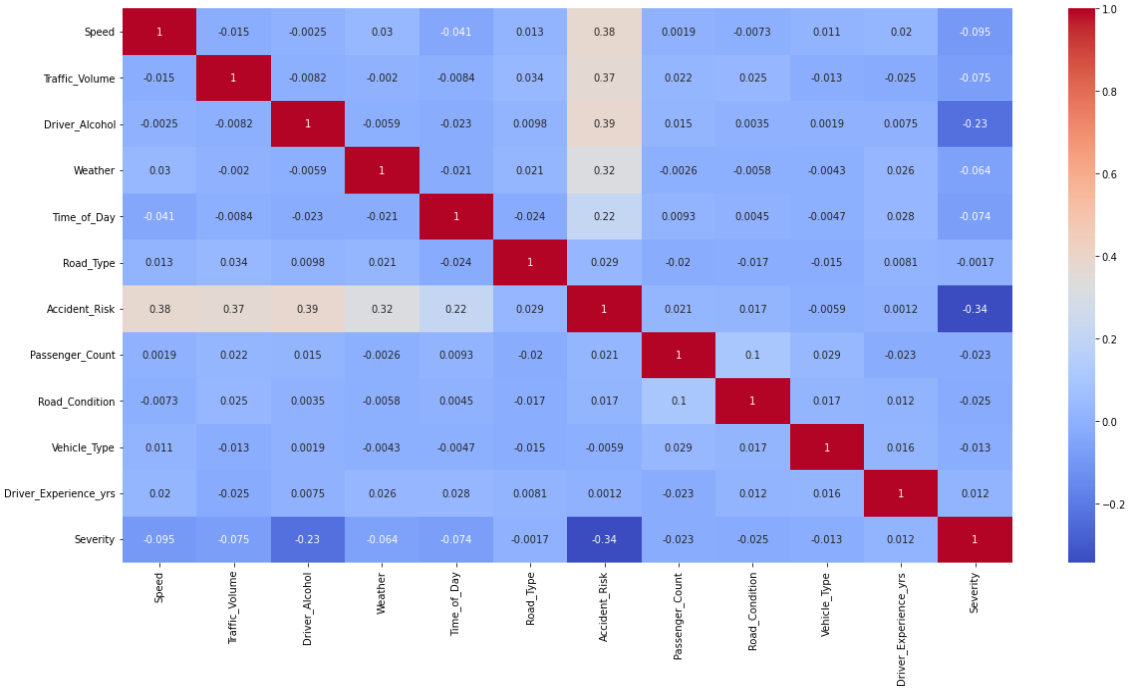

Fig.2 Heatmap showing corelation.

```
-- Logistic Regression --
              precision    recall  f1-score   support

           0       1.00      0.99      0.99       148
           1       0.94      0.97      0.96       247
           2       0.95      0.93      0.94       205

    accuracy                           0.96       600
   macro avg       0.97      0.96      0.96       600
weighted avg       0.96      0.96      0.96       600

Accuracy: 0.9616666666666667
F1 Score: 0.9616632142618267


-- SVM --
              precision    recall  f1-score   support

           0       0.97      0.95      0.96       148
           1       0.95      0.97      0.96       247
           2       0.93      0.92      0.92       205

    accuracy                           0.95       600
   macro avg       0.95      0.95      0.95       600
weighted avg       0.95      0.95      0.95       600

Accuracy: 0.9483333333333334
F1 Score: 0.9483101833024218


-- KNN --
              precision    recall  f1-score   support

           0       0.81      0.72      0.76       148
           1       0.86      0.88      0.87       247
           2       0.68      0.71      0.69       205

    accuracy                           0.78       600
   macro avg       0.78      0.77      0.78       600
weighted avg       0.78      0.78      0.78       600

Accuracy: 0.7833333333333333
F1 Score: 0.7832563650968322


=== Random Forest ===
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       148
           1       1.00      1.00      1.00       247
           2       1.00      1.00      1.00       205

    accuracy                           1.00       600
   macro avg       1.00      1.00      1.00       600
weighted avg       1.00      1.00      1.00       600

Accuracy: 1.0
F1 Score: 1.0
```

6. References:

Books:
1. Let Us Python – Yashavant Kanetkar

Online Resources:
• GeeksforGeeks – https://www.geeksforgeeks.org/
• TutorialsPoint – https://www.tutorialspoint.com/cprogramming/index.htm
• Stack Overflow – https://stackoverflow.com